

Rahul Rawat

Mandatory Intern, Data Science and Deep Learning for Energy Systems

PERSONAL DETAILS

Address	C2 16, 68159 Mannheim, Germany
Phone	015563603340
Email	rahulrawat2r@gmail.com
LinkedIn	linkedin.com/in/rahulrawat2r
GitHub	github.com/rahulrawat20022002
Portfolio	rah-portfolio.pages.dev
Date of birth	20 February 2002
Nationality	Indian, student visa with valid work permit
Availability	Werkstudent 20 hours per week immediately, full time from April 2027

PROFILE

Data Science and Analytics Master student at SRH Heidelberg based in Mannheim, with hands on time series forecasting, deep learning, and cloud data pipeline work on real sensor data. I have shipped a recursive time series pipeline at eRay GmbH forecasting four water quality indicators for a German lake using CatBoost multi quantile regression with asymmetric **80 percent** prediction intervals, a real time Google Cloud pipeline processing over **128 thousand** flight records enriched against four data sources every **30 seconds**, and a fairness by design classification system with rigorous evaluation and unit tests at **100 percent** branch coverage. Proficient in Python and deep learning frameworks, comfortable with anti leakage evaluation, I am the right fit for a Mandatory Internship on Data Science and Deep Learning for Energy Systems at Siemens.

PROFESSIONAL EXPERIENCE

Oct 2025 to Mar 2026 Heidelberg	eRay GmbH Data Scientist - Built an end to end recursive time series pipeline forecasting chlorophyll a, turbidity, pH, and dissolved oxygen for a German lake, delivered as a six month collaboration with SRH University Heidelberg. - Benchmarked six models head to head, Ridge, Gradient Boosting, LightGBM, XGBoost, CatBoost, and Prophet, and used CatBoost multi quantile regression to produce asymmetric 80 percent prediction intervals for decision support under uncertainty. - Enforced strict anti leakage rules across the pipeline, surfacing the honest finding that pH and dissolved oxygen are physically predictable while chlorophyll a and turbidity are not without live optical sensors. - Reconstructed missing winter readings with MICE imputation and engineered a synthetic winter decay forecast canvas so tree based models stopped flatlining during recursive prediction. - Wrapped the whole pipeline in an orchestrator with gate checks and velocity and ecological bounds that halts on failed imputation rather than letting bad data cascade downstream.
------------------------------------	--

Technologies: *Python, CatBoost, Prophet, scikit learn, MICE*

PERSONAL PROJECTS

2025

Real Time Flight Tracking Data Pipeline

- Collected live flight positions from the OpenSky Network API every **30 seconds** and enriched them against airport, aircraft, and weather data across four sources, measured by a clean joined table covering more than **128 thousand** records, using Python collectors and PySpark cleaning on Google Cloud.
- Shaped raw collector output into analysis ready tables with dbt and computed each aircraft's nearest airport along the way, measured by consistent nearest airport labels across the historical dataset, using PySpark for heavy lift and dbt for the modelling layer.
- Orchestrated the whole system with Apache Airflow so batch and real time layers refresh automatically every **15 minutes**, measured by unattended DAG runs on GCS backed storage and Dataproc compute.
- Surfaced findings in a Tableau workbook backed by Python statistics through TabPy, measured by the finding that air traffic drops **4.4 times** in heavy rain and clusters around hubs like Frankfurt and Munich, using dbt aggregates as the visual layer feed.

Technologies: *Python, PySpark, BigQuery, dbt, Apache Airflow, GCP Dataproc and GCS, Tableau with TabPy*

2025

CreditIQ: Fairness by Design Credit Scoring

- Raised the model's Disparate Impact ratio from a failing **0.79** to a compliant **0.88**, measured against the EU AI Act and AGG **80 percent** fairness threshold, using AIF360 mitigation and threshold calibration on a real credit dataset.
- Diagnosed a hidden intersectional bias where younger women were still penalised even after single axis age bias was fixed, measured through SHAP driven subgroup analysis, then designed a four way age by gender threshold matrix to correct it without over correcting into reverse discrimination.
- Cut the false negative rate from **44 percent** to **16.7 percent** while holding accuracy at **75 percent**, measured on the held out test split, by documenting the fairness accuracy trade off as a deliberate and regulator defensible decision rather than a hidden compromise.
- Shipped a Streamlit decision support tool that gives finance managers a recommendation plus a plain language LLM generated explanation, measured against GDPR Article 22 and EU AI Act Article 14 human in the loop requirements, and backed the pipeline with unit tests at **100 percent** branch coverage and a full regulatory write up.

Technologies: *Python, scikit learn, AIF360, SHAP, Streamlit*

2025

Movie Analytics and ML Pipeline on GCP

- Built an end to end batch pipeline pulling movie data from a public API into a GCS data lake and processing it through Bronze to Silver to Gold medallion architecture in BigQuery, measured by a fully automated Cloud Scheduler trigger with no manual intervention required, using Cloud Run for compute.
- Engineered the Silver layer for trustworthy analytics with schema enforcement, safe type casting, deduplication via window functions, and genre normalisation into a relational model, measured by clean referential integrity across all downstream Gold tables.
- Trained a BigQuery ML classifier that predicts whether a film will be a hit before release, measured by leakage free evaluation, by deliberately splitting features into two tables so only pre release signals feed the model.
- Built Gold layer aggregates and a five page Looker Studio dashboard answering concrete business questions on genre ROI, foreign language growth, and release season timing, measured against the ML early warning view, and secured the system with a least privilege service account and Secret Manager.

Technologies: *GCP BigQuery, Cloud Run, GCS, Cloud Scheduler, BigQuery ML, Python, SQL, Looker Studio*

EDUCATION

Apr 2025 to Present Heidelberg	M.Sc. Data Science and Analytics SRH University of Applied Sciences Heidelberg, GPA 1.9
2019 to 2023 Greater Noida, India	Bachelor of Technology in Computer Science GL Bajaj Institute of Technology and Management, CGPA 7.3 of 10

RESEARCH AND THESIS

2022 to 2023 Greater Noida, India	Bachelor Thesis, Diabetes Prediction Using Machine Learning GL Bajaj Institute of Technology and Management, IEEE style paper - Built a full end to end machine learning pipeline comparing six classifiers on a clinical dataset of 768 patients, using 10 fold cross validation and per model confusion matrices to make the model comparison auditable. - Caught biologically impossible zero values that the original authors had overlooked, then applied IQR based outlier removal and proper imputation to restore data integrity before any model fit. - Chose ROC AUC over accuracy as the headline metric for a 65 to 35 imbalanced dataset, avoiding a misleadingly rosy accuracy figure that would have hidden real error patterns in the minority class. - Wrote up findings as an IEEE style paper including an honest section on limitations and what to do differently in a follow up study, publishable in substance. Technologies: <i>Python, scikit learn, Pandas, Seaborn</i>
--------------------------------------	---

CERTIFICATIONS

- NVIDIA: Building LLM Applications With Prompt Engineering, issued November 2025.
- AWS Academy Graduate: AWS Academy Cloud Foundations, issued July 2025.
- SAS Certified Specialist: Visual Business Analytics Using SAS Viya, issued May 2025.

ACHIEVEMENTS

- USAII Global AI Hackathon 2026: Finalist at Graduate Level, awarded by the United States Artificial Intelligence Institute for innovation, technical creativity, and applied AI on real world challenges.

LANGUAGES

English	Fluent, written and spoken
German	B1 in progress toward B2
Hindi	Native

Mannheim, 22 July 2026
Rahul Rawat